

Terraform VPC - LAB

```
provider "aws" {  
  region    = "us-east-2"  
  access_key = ""  
  secret_key = ""  
}
```

```
#####  
# VPC  
#####  
resource "aws_vpc" "main" {  
  cidr_block      = "10.0.0.0/16"  
  enable_dns_support = true  
  enable_dns_hostnames = true  
  
  tags = {  
    Name = "complete-vpc"  
  }  
}
```

```
#####  
# Internet Gateway  
#####  
resource "aws_internet_gateway" "igw" {  
  vpc_id = aws_vpc.main.id  
  
  tags = {  
    Name = "complete-igw"  
  }  
}
```

```
#####  
# Public Subnet  
#####  
resource "aws_subnet" "public" {
```

```

vpc_id      = aws_vpc.main.id
cidr_block  = "10.0.1.0/24"
availability_zone = "us-east-2a"
map_public_ip_on_launch = true

tags = {
  Name = "complete-public-subnet"
}
}

#####
# Private Subnet
#####
resource "aws_subnet" "private" {
  vpc_id      = aws_vpc.main.id
  cidr_block  = "10.0.2.0/24"
  availability_zone = "us-east-2a"

  tags = {
    Name = "complete-private-subnet"
  }
}

#####
# Public Route Table
#####
resource "aws_route_table" "public_rt" {
  vpc_id = aws_vpc.main.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }

  tags = {
    Name = "complete-public-rt"
  }
}

```

```
}
```

```
resource "aws_route_table_association" "public_assoc" {  
  subnet_id    = aws_subnet.public.id  
  route_table_id = aws_route_table.public_rt.id  
}
```

```
#####
```

```
# Elastic IP for NAT Gateway (no vpc = true required)
```

```
#####
```

```
resource "aws_eip" "nat_eip" {  
  tags = {  
    Name = "complete-nat-eip"  
  }  
}
```

```
#####
```

```
# NAT Gateway
```

```
#####
```

```
resource "aws_nat_gateway" "nat" {  
  allocation_id = aws_eip.nat_eip.id  
  subnet_id    = aws_subnet.public.id
```

```
  depends_on = [aws_internet_gateway.igw]
```

```
  tags = {  
    Name = "complete-nat-gateway"  
  }  
}
```

```
#####
```

```
# Private Route Table
```

```
#####
```

```
resource "aws_route_table" "private_rt" {  
  vpc_id = aws_vpc.main.id
```

```
  route {
```

```
cidr_block    = "0.0.0.0/0"
nat_gateway_id = aws_nat_gateway.nat.id
}
```

```
tags = {
  Name = "complete-private-rt"
}
}
```

```
resource "aws_route_table_association" "private_assoc" {
  subnet_id    = aws_subnet.private.id
  route_table_id = aws_route_table.private_rt.id
}
```